IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

COMMUNICATION NETWORKS

Friday, 15 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J.A. Barria
Second Marker(s) : C. Ling

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Information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,
$1/\mu$ = Average length of packet [bit/packet]
C_i = Transmission speed link *i* [bits/s]
μC_i = Service rate (link *i*) [packet/ s]
λ_i = Arrival rate (link *i*) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{ F_i \}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.
F_i = Flow carried by link l_i.

3. Velocity of propagation and propagation delay.

Velocity of unguided transmission through air or space of an electromagnetic wave is $\mathbf{2.99x10^8}$ m/s

Assume velocity of guided transmission to be approximately $\mathbf{66}$% of the unguided wave propagation.

And propagation delay:

$$\boldsymbol{a = \frac{Rd}{LV}}$$

Where,
$\boldsymbol{R}$ = Data rate of the channel.
$\boldsymbol{d}$ = Length of the network.
$\boldsymbol{L}$ = Average frame length.
$\boldsymbol{V}$ = Velocity of guided media.

The Questions

Q1.

a)

For a sliding window flow control, at the link layer, more than one frame can be transmitted without acknowledgment from the receiver end.

Define parameter a as:

$$a = \frac{propagation\ delay}{trasmission\ delay}$$

And,

- assume that the transmission delay is normalized, hence a represents the end-to-end propagation delay.
- assume that the probability of a single frame is in error is known and represented by the parameter P.

i) Consider W as the maximum number of frames allowed in the media (i.e. is the size of the window). Estimate the number of outstanding frames when the window size $W \geq 2a + 1$ and when the window size $W < 2a + 1$. Clearly elaborate your arguments. **[4]**

ii) Estimate the utilisation of a **Go Back N ARQ** mechanism for the complete range of window sizes. Show the derivation step by step including all assumptions made. **[7]**

b)

i) Simple Network Management Protocol (SNMP): Describe SNMP purpose, it mode of messaging and discuss on top of which TCP/IP layer this application runs and why. **[3]**

ii) Internet Control Message Protocol (ICMP) protocol: Describe the purpose of this protocol and how ICMP messages are carried. **[3]**

iii) Discuss the purpose of a ready to send (RTS) and clear to send (CTS) exchange of messages in the IEEE 802.11 MAC protocol. Specifically discuss how relevant it is when the data that is being transported is of the same order of magnitude to the size the exchange of messages. **[3]**

Q2.

a) Discuss the similarities and differences between the following type of routing algorithms: i) link-state routing vs ii) distance-vector routing. **[4]**

For the routing protocols BGP, RIP and OSPF: classify them as interior or exterior gateway protocol, identify which type of routing algorithm they use, and what TCP/IP suite protocol layer they rely on. **[4]**

b) The total power consumption on a wireless link is given by the following expression:

$$\text{Power consumed} = (E_t + E_r)B$$

where the energy/bit consumed: $E_t = e_t + e_d d^3$ and $E_r = e_r$. The values of the parameters are: $e_t = e_r = 5 \cdot 10^{-10}$ Jules/bit. $e_d = 1 \cdot 10^{-10}$ Jules/bit/m^3.

$d =$ distance in metres between communicating nodes.

i) Using as a reference **Fig 2.1.** node **M** is moving away from node **1** and approaching node **2** in a straight line. Knowing that **d** = 20 m derive analytically the distance **x** from node **1** where the power consumption is minimised. Assume $B = 2$ MBit/s. **[6]**

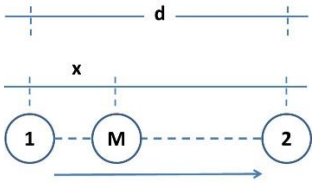


Figure 2.1: three nodes wireless network **d** = 20 m.

ii) For the network shown in **Fig. 2.2.** and considering a simplified expression for power consumption, $e_d d^3$, identify the lowest power consumption path of the O-D communication between nodes **1** and **6**. Assume $B = 4$ MBit/s. Explicitly derive the cost of each hop. **[3]**

iii) Discuss if by moving node **5** to another position you would be able to further minimise the power consumption of the O-D communication between nodes **1** and **6**. Assume $B = 4$ MBit/s. Clearly argue your solution. **[3]**

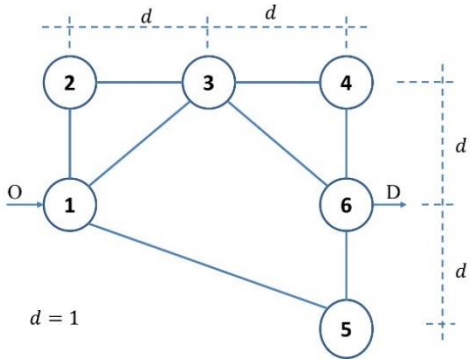


Figure 2.2: Wireless network, where $d = 1$ m.

Q3.

a)

i) Describe and briefly discuss the difference between scheduling policy and queue management. **[3]**

ii) Identify and discuss how a parameter of a leaky bucket algorithm can control the maximum mean rate of transmitted packets over the long run. **[3]**

iii) Discuss how a leaky bucket algorithm controls the size of a burst of transmitted packets. **[3]**

iv) Define jitter and discuss how you could smooth-out the jitter of arriving packets at a receiver node in a multi-media network. **[3]**

b)

Using the network shown in **Fig. 3.1.** allocate flows using a Network Utility Maximisation (NUM) objective function:

$$NUM_{obj} = \sum_i \frac{w(i)^{\alpha} r(i)^{1-\alpha}}{(1-\alpha)}$$

Note: Assume that $r(a) + r(b) = r(2) = r(3)$ and for this problem can be regarded as an aggregated flow $r(4)$.

i) Formally state the Weighted Fair flow allocation: for $w(i) \neq 1$ and $\alpha \rightarrow 1$. **[3]**

ii) Prove that the flow allocation in i) is a generalisation of the Proportional Fair allocation, when $w(i) = 1$ and $\alpha \rightarrow 1$**.** **[2]**

iii) Prove that the flow allocation in i) can implement a Max-min fair flow allocation for the network flow allocation problem shown in **Fig. 3.1**. **[3]**

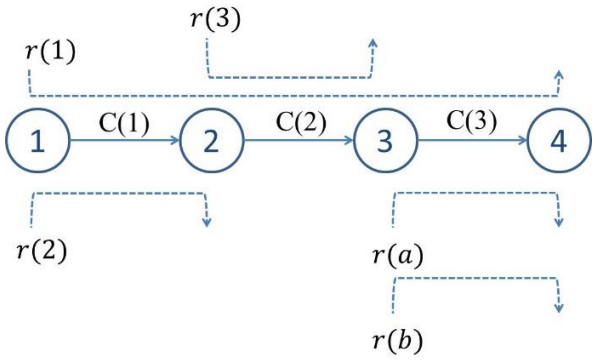


Figure 3.1: NUM flow allocation problem.

Q.4.

a)

i) Briefly discuss in your own words the need to implement the TCP congestion control states: how slow start works and how it differs from a congestion avoidance state? **[5]**

ii) Briefly discuss what is meant by additive increase / multiplicative decrease in relation to a TCP congestion control mechanism. **[2]**

b)

For the packet network in **Fig. 4.1** all external arrivals are assumed Poisson streams with mean arrival rate γ_{ij} packets per second. The traffic matrix T is given:

$$T = \begin{bmatrix} 0 & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & 0 & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0.35 & 0.25 \\ 0.35 & 0 & 0.15 \\ 0.25 & 0.15 & 0 \end{bmatrix}$$

All links have capacity: $C(i,j)$ = 4 Kbits per second.

Packet lengths are exponentially distributed with mean length 6 Kbits/packet.

i) Derive the mean number of outstanding packets N in the network. **[4]**

ii) You are permitted to add an additional bidirectional connection between two pair of nodes: Identify your chosen connection and discuss the reason for this choice. **[3]**

iii) Solve the routing problem using min-hop count as cost function. Identify the bottleneck link of the resulting network. **[3]**

iv) Derive the mean number of outstanding packets N' in the resulting network. Contrast N with N'. **[3]**

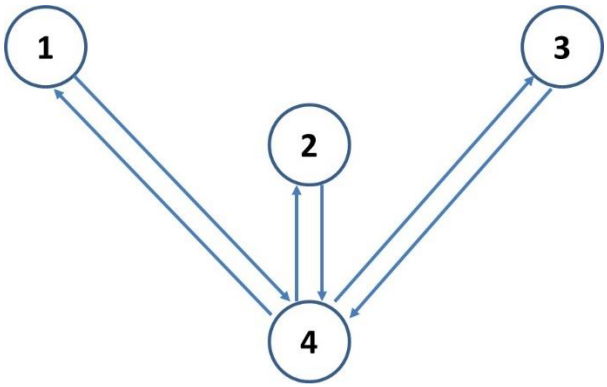


Figure 4.1.: Four node network.